# Xinwei Zhuang

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## Research Emphasis

**Energy and Environment:** AI for sustainable energy transition; Building stock analysis; Urban building energy modeling; Energy conservation and projection; Ventilation and daylighting optimization

**Computation:** Machine learning; Data science; Network science; Computer vision; Computer graphics

## Education

**Massachusetts Institute of Technology** — Cambridge, United States
IvyPlus Exchange Scholar, Department of Urban Studies and Planning

**University of California, Berkeley** — Berkeley, United States
Ph.D. in Architecture

**The Bartlett School of Architecture, University College London** — London, United Kingdom
M.S. in Architectural Computation

**Edinburgh Napier University** — Edinburgh, United Kingdom
B.S. in Civil Engineering

## Experience

**Texas A&M University** — College Station, United States
*Assistant Professorr* — 2025 – Present

**Lawrence Berkeley National Lab** — Berkeley, United States
*Research assistant* — 2022 – 2025

- Decarbonizing Energy through Collaborative Analysis Routes and Benefits (DECARB)
- Multi-scale Energy Conservation Measurement (ECM) evaluation and projection for U.S. residential and commercial building sectors, provide policy insight for ECMs and regulations

**Architecture Studio** — Shanghai, China
*Architectural designer* — 2018 – 2020

**China Shanghai Architectural Design & Research Institute Co. Ltd** — Shanghai, China
*Research assistant* — 2017 – 2018

## Invited Talk

**2026** Block-Scale Solar PV and Battery Storage Co-Optimization for Energy Resilience
*ASHREA summer conference, Austin, TX, USA*

**2026** Community-driven energy resilience with decentralized networks
*COA Showcase, Texas A&M University, TX, USA*

**2025** When the city goes dark: Measuring building vacancy based on energy consumption in Rio de Janeiro
*Presentation at MIT Energy Conference 2025, Boston, MA, USA*

**2024** The role of architecture in the urban energy landscape
*Guest lecture at the Manchester Urban Institute, University of Manchester, UK*

**2024** Decomposing and recomposing of the urban energy environment with building archetype and urban building energy modeling
*Guest lecture at Foster + Partner, UK*

**2023** Performance-driven architectural design with deep learning
*Invited talk at 33rd Young Scholar Forum, Nanjing University, China*

## Grants

**2025** Co-PI Targeted Proposal Teams (**$50k**)
*Multi-Source Integration of New Satellites, Low-Cost Sensors, and Human Mobility for Precision Air Pollution Exposure Assessment*

**2024** Co-PI Lau Grants for Just Climate Futures (**$35k**)
*Energy-Sharing Neighborhood: Community-Driven Energy Resilience with Decentralized Networks*

**2024** PI Hearts to Humanity Eternal (**$10k**)
*Urban Synergy: Performance-Driven Urban Neighborhood Modeling with Hierarchical Networks for Energy Resilience*

## Selected Publication

* equal contribution

1. **Zhuang X.** and Tan H. (2026) Automated Neighborhood Archetype: Integrating Morphology and Topology for Urban Building Energy Modeling, *Energy and Buildings*. Vol. 362, Article 117504. DOI: 10.1016/j.enbuild.2026.117504

2. **Zhuang X.**, Lyv G., Zhao Z. and Caldas. L (2025) Rapid Assessment of Solar Potential for Building Surfaces in Complex Urban Morphologies Based on Vector Processing, *Solar Energy*, Volume 294, 2025, 113482, ISSN 0038-092X, DOI: 10.1016/j.solener.2025.113482

3. **Zhuang X.**, Zhu P., Yang A. and Caldas L. (2025) Machine Learning for Generative Architectural Design: Advancements, Opportunities, and Challenges, *Automation in Construction*, Volume 174, 2025, 106129, ISSN 0926-5805, DOI: 10.1016/j.autcon.2025.106129.

4. Chen X., Lyv G., **Zhuang X.**, Duarte C. and Schiavon S. (2024) Integrating Symbolic Neural Networks for Advanced Modeling in Building Physics: A Study and Proposal. https://arxiv.org/abs/2411.00800

5. **Zhuang X.**, Chu X., Liang, J., Zhu P., Gonzalez, M. and Caldas, L. (2024) Across scales: Hierarchical Urban Graph for Neighborhoods Partition and Decentralized Energy Autonomy, in *2024 Conference on Association for Computer Aided Design in Architecture (ACADIA)*, Calgary, Canada. [paper]

6. **Zhuang X.***, Huang Z.*, Zeng W., and Caldas L. (2023) MARL: Multi-scale Archetype Representation Learning for Urban Building Energy Modeling, workshop on Computer Vision Aided Architectural Design (CVAAD), at *International Conference on Computer Vision (ICCV)*, Paris, France. DOI: 10.1109/ICCVW60793.2023.00171

7. **Zhuang X.***, Huang Z.*, Zeng W., and Caldas L. (2023) Encoding Urban Ecologies: Automated Building Archetype Generation through Self-Supervised Learning for Energy Modeling, in *2023 Conference on Association for Computer Aided Design in Architecture (ACADIA)*, Denver, United States of America. papers.cumincad.org/cgi-bin/works/paper/acadia23_v2_532

8. **Zhuang X.**, Luo, N., Hong T. and Koenig, M. (2023) What can we learn from Honda Smart Home with high-resolution monitored performance data: A zero-net energy home in California, in *18th International IBPSA Conference and Exhibition* In Shanghai, China. DOI: 10.26868/25222708.2023.1315

9. Li F.* and **Zhuang X.*** (2023) Evaluating an Auto Decoder-based Generative Model for the Infomorphism Urban Planning Framework, in *18th International IBPSA Conference and Exhibition* In Shanghai, China. DOI: 10.26868/25222708.2023.1393

10. **Zhuang X.**, Ju Y., Yang A. and Caldas L. (2023) Synthesis and Generation for 3D Architecture Volume with Generative Modeling, in *International Journal of Architectural Computing*, Vol. 21, Issue 2: AI, Architecture, Accessibility, & Data Justice. DOI: 10.1177/14780771231168233